

Using a Microeconometric Model of Household Labour Supply to Design Optimal Income Taxes*

Rolf Aaberge

Statistics Norway, NO-0033 Oslo, Norway
rolf.aaberge@ssb.no

Ugo Colombino

University of Torino, 10153 Torino, Italy
ugo.colombino@unito.it

Abstract

With regards to empirical applications of optimal taxation theory, analytical expressions are typically adopted for optimal taxes, and then numerical values are imputed to their parameters by calibration or by using previous estimates. We aim to avoid the restrictive assumptions and possible inconsistencies of this approach. In contrast, we identify optimal taxes by iteratively running a microeconometric model, based on 1994 Norwegian data, until a given social welfare function is maximized, given the public budget constraint. The optimal rules envisage monotonically increasing marginal rates (negative on very low incomes) and—compared to the current rule—a lower average rate, lower marginal rates on low incomes, and higher marginal rates on very high incomes.

Keywords: Microsimulation; optimal taxation; random utility model

JEL classification: H21; H31; J22

I. Introduction

In this paper, we present an empirical analysis of optimal income taxation. The purpose is not new, but the exercise illustrated here differs, in many important ways, from previous attempts to compute optimal taxes empirically. The standard procedure adopted in the literature starts with some version of the optimal taxation framework, originally set up in the influential paper by Mirrlees (1971). The next step typically consists of imputing numerical values—either determined by calibration or taken from previous empirical analysis—to the parameters (e.g., wage elasticities of labour

*We thank Tom Wennemo for skilful programming assistance, and we are grateful to a referee for very detailed and useful comments and suggestions. U. Colombino received financial support from the Italian Ministry of University and Research and the Compagnia di San Paolo. ICER is gratefully acknowledged for providing financial support and excellent working conditions for R. Aaberge.

supply) appearing in the formulae produced by the theory. This body of literature has been surveyed by Tuomala (1990). A recent strand of research adopts a similar approach to address the inverse optimal taxation problem (i.e., retrieving the social welfare function that makes optimal a given tax rule; Bourguignon and Spadaro, 2005). There are two main problems with the optimal taxation literature, as follows. (a) The theoretical results become amenable to an operational interpretation only by adopting rather restrictive assumptions concerning the preferences, the composition of the population, and the structure of the tax rule. (b) The empirical measures used as counterparts of the theoretical concepts are usually derived from previous estimates obtained under assumptions that are different from those used in the theoretical model. As a consequence, the consistency between the theoretical model and the empirical measures is dubious, and the significance of the numerical results remains uncertain. The typical outcome of these exercises envisages a lump-sum transfer, which is progressively taxed away by very high marginal tax rates (MTRs) on lower incomes (i.e., a negative income tax mechanism). Beyond the “break-even point” (i.e., the income level where the transfer is completely exhausted), the MTRs are close to constant. However, Tuomala (2010) has suggested that these results are essentially forced by the restrictive assumptions made upon preferences, labour supply elasticities, and the distribution of productivities (or wage rates). Interestingly, when Tuomala (2010) adopts a more flexible specification of the utility function, he finds that the optimal system is progressive, with monotonically increasing MTRs.

While most of the studies mentioned above have been essentially illustrative numerical exercises, several recent contributions have attempted to use optimal taxation results in the empirical evaluation or design of tax-transfer reforms. Diamond (1980, 1998), Revesz (1989), and Saez (2001) have made the results of Mirrlees easier to interpret by reformulating them in terms of labour (or income) supply elasticities, in order to provide a direct link between the theoretical results and empirical measures. Saez (2002) has developed a model that is amenable to empirical implementation, and that focuses on the relative magnitude of the labour supply elasticities at the extensive and intensive margins. Immervoll *et al.* (2007) have adopted the approach of Saez (2002) to evaluate alternative income support policies in European countries. Blundell *et al.* (2009) and Haan and Wrohlich (2010) have also used the approach of Saez (2002) to evaluate taxes and transfers for lone mothers in Germany and the UK, whereas Kleven *et al.* (2009) have provided results on the taxation of couples. Although these new contributions are interesting attempts in the direction of an empirical implementation of theoretical optimal taxation results, they still rely on restrictive assumptions and suffer from a possible inconsistency between the theoretical model and the empirical measures

used to implement it. For example, the basic version of the model proposed by Saez (2002), which has been adopted in the empirical exercises mentioned above, does not account for income effects.¹ Moreover, it relies on rather restrictive assumptions upon the way the households respond to changes in the relative attractiveness of the opportunities in the budget set.² When it comes to empirical applications (as in Immervoll *et al.*, 2007; Blundell *et al.*, 2009; Haan and Wrohlich, 2010), the parameters of the theoretical models are given numerical values estimated with microeconomic models that do not adopt the same restrictive assumptions as in Saez (2002). Of course, some of these limitations and potential inconsistencies might be overcome in the future, but analytical solutions of the optimal income taxation problem will likely never be fully consistent with flexible structural labour supply models. Here, we follow a different, and possibly complementary, approach. We do not start from theoretical results that dictate the conditions for optimal tax rules under various assumptions. Instead, we use a microeconomic model of labour supply in order to identify, by simulation, the tax rule that maximizes a social welfare function, under the constraints that the households maximize their own utility and that the total net tax revenue remains constant. The microeconomic simulation approach is commonly used to evaluate tax reforms, but it has not often been used in empirical optimal taxation studies. The closest examples in which a similar approach is adopted are Fortin *et al.* (1993), Aaberge and Colombino (2006), Colombino *et al.* (2010), and Blundell and Shephard (2012).³ Analytical solutions are still crucial for understanding the “grammar” of the problem, and for suggesting promising directions of reform. In contrast, microeconomic models and computational solutions allow for the introduction of less restrictive specifications of preferences and opportunity sets, and for the evaluation of more complex tax-transfer rules. The estimated model we use here represents the choices of both couples and singles, it adopts a flexible specification of the preferences, it accounts for quantity constraints in labour supply choices, and it can

¹Income effects can be accounted for, as in Saez (2001), at the cost of notable analytical and computational complications.

²In Saez (2002), each individual has only three opportunities from which to choose: non-participation, and two adjacent labour income brackets.

³Fortin *et al.* (1993) have used a calibrated (not estimated) model with rather restrictive (Stone–Geary) preferences, and have focused on alternative income support schemes rather than on the whole tax rule. Aaberge and Colombino (2006) have reported on the preliminary results of a simpler version of the exercise illustrated in this paper. Colombino *et al.* (2010) have analysed the basic income support mechanisms in some European countries. Blundell and Shephard (2012) have identified the optimal design of a specific UK policy that addresses low-income families with children. However, they have not treated the problem of interpersonal comparability, which, in their case, might be less important, given the smaller and less heterogeneous target population.

accommodate a detailed representation of complex tax-transfer systems. The optimal tax rules turn out to envisage an average tax rate lower than the current one, a modest lump-sum tax (interpretable as a property tax), a negative tax on low incomes (close to mechanisms such as the Earned Income Tax Credit or the In-Work Tax Credit policies), and a progressive MTR profile culminating in a 100 per cent MRT on very high incomes (about 1.5 per cent of the sample). This scenario contrasts sharply with the results obtained by the numerical exercises inspired by the leading contribution of Mirrlees (1971). It is closer to the picture that is typically revealed by empirical applications that adopt the theoretical results of Saez (2002). However, using a flexible microeconomic model as a computational tool, we are able to explore a larger variety of tax-transfer rules, and to perform a more articulated analysis of the effects of the various rules upon different segments of the population. Obviously, the results of our computational exercise cannot claim a generality that is similar to the analytical solutions. While analytical solutions establish an explicit relationship between the fundamentals of the economy (preferences, skill distribution, etc.), the results of our computational exercise are application-specific (in this paper, Norway-specific); this is the price of taking into account a more detailed and flexible representation of preferences and opportunity sets. However, in principle, this limitation of our computational exercise can be overcome. By performing similar exercises on many different economies, it should again be possible to identify—empirically—a more general relationship between the fundamentals of the economy and the optimal income tax rule.

As explained in Section II, the microeconomic model used in this study contains 78 parameters that capture the heterogeneity of preferences and opportunities among households and individuals. The estimated model is used to simulate the choices, given a particular tax rule. Thus, these choices are generated by preferences and opportunities that vary across the decision units. However, because preferences are heterogeneous and because some individuals are single, whereas others form families and live together, when it comes to social evaluation it does not make sense to treat the estimated utility functions as comparable individual welfare functions. In order to solve the interpersonal comparability problem, we adopt a method that consists of using a common utility function in order to produce interpersonally comparable individual welfare measures to be used as arguments of the social welfare function. The common utility function is justified as a normative standard where the social planner treats individuals symmetrically, and it is only used to compute and compare the individual welfare levels that provide the basis for the social welfare evaluation of tax reforms; it is not used for simulating household behaviour (where, instead, the estimated individual utility functions are used). This procedure, which circumvents the problem of interpersonal comparability of

heterogeneous preferences, is well established in the body of the literature concerning empirical public economics. It has been proposed by Deaton and Muellbauer (1980) and Hammond (1991), and it forms the basis for the definition and measurement of a money-metric measure of utility in King (1983) and Aaberge *et al.* (2004). Moreover, it has been applied, for example, by Fortin *et al.* (1993), Colombino (1998), and Colombino *et al.* (2010). As a practical matter, an average of the estimated individual utility functions or an estimated utility function (individual welfare function) with common parameters (as in our case) is typically used.

We present a brief description of the microeconomic model in Section II. The empirical specification of the model and the estimates of the model parameters have already been provided by Aaberge and Colombino (2011). In order to illustrate the behavioural implications of the estimates, in Section II, we report the wage elasticities of labour supply, whereas Aaberge and Colombino (2011) have presented the income elasticities. Because the microeconomic model, once estimated, is used for a rather ambitious purpose (i.e., simulating choices with a view to identifying optimal tax rules), it is important to check its reliability. Ultimately, the model should be judged on its ability to carry out the job it has been built for (i.e., predicting the outcomes of policy changes). In Section II, therefore, we perform an out-of-sample prediction exercise where we use the model (estimated on 1994 data) to predict household-specific distributions of income in Norway in 2001. In Section III, we present the social welfare evaluation method and the computational procedure for solving the optimal taxation problem. The resulting optimal rules are presented in Section IV. In Section V, we provide some final comments.

II. Modelling Framework

Microeconomic Labour Supply Model

In this section, we present a sketch of the microeconomic model. A full description has been given by Aaberge and Colombino (2011). The model can be considered as an extension of the standard multinomial logit model, and it differs from the traditional models of labour supply in several respects.⁴ First, it accounts for observed as well as unobserved heterogeneity in preferences and constraints. Second, it includes both single-person households and married or cohabiting couples who make joint labour-supply decisions. A proper model of the interaction between spouses in their labour-supply decisions is important, because most of the individuals

⁴Examples of previous applications of this approach can be found in Aaberge *et al.* (1995, 1999, 2000, 2004).

are married or cohabiting. Third, by taking all the details of the tax system into account, the budget sets become complex and non-convex in certain intervals. For expository simplicity, in this section we consider only the behaviour of a single-person household. The extension to couples has been fully explained by Aaberge and Colombino (2011). The agents choose a job from alternatives characterized by the wage rate w , hours of work h , and other characteristics j . The problem solved by the agent is

$$\begin{aligned} & \max_{(w,h,s,j) \in B} U(c, h, s, j) \\ & \text{s.t.} \\ & c = f(wh, I) \end{aligned} \quad (1)$$

where h is hours of work, w is the pre-tax wage rate, s are the observed job characteristics (besides h and w ; e.g., occupational sector), j are the unobserved (by the analyst) job and/or household characteristics, I is the pre-tax non-labour income (exogenous), c is the net disposable income, f is the tax rule that transforms gross pre-tax incomes (wh , I) into net disposable income c , and B is the set of all opportunities available to the household (including non-market opportunities, or “leisure” activities; i.e., “jobs” with $w = 0$ and $h = 0$). Agents can differ not only in their preferences and in their wage (as in the traditional model) but also in the number of available jobs of different types. Moreover, for the same agent, wage rates (unlike in the traditional model) can differ from job to job. Let $p(h, w, s)$ denote the density of available jobs of type (h, w, s) . By representing the choice set B by a probability density p , we can, for example, allow for the fact that jobs with hours of work in a certain range are more or less likely to be found, possibly depending on agents’ characteristics, or for the fact that, for different agents, the relative number of market opportunities might differ. We assume that the utility function can be factorized as

$$U[f(wh, I), h, s, j] = v[f(wh, I), h, s] \varepsilon(j), \quad (2)$$

where v and $\varepsilon(j)$ are the systematic and random components, respectively. The term $\varepsilon(j)$ is a random variable, which accounts for the effect on utility of all the characteristics of the household–job match that are observed by the household but not by us. Moreover, we assume that $\varepsilon(j)$ is independently and identically distributed, according to the type III extreme value distribution. We observe the chosen h , w , and s . Therefore, we can specify the probability that the agent chooses a job with observed characteristics (h, w, s) . It can be shown that, under assumptions (1) and (2), and given the extreme value distribution for ε , we can write the probability density

function of a choice (h, w, s) as⁵

$$\varphi(h, w, s) = \frac{v[f(wh, I), h, s] p(h, w, s)}{\int \int \int v[f(xy, I), y, z] p(y, x, z) dx dy dz}. \quad (3)$$

The density (3) is the contribution of an observation (h, w, s) to the likelihood function, which is then maximized in order to estimate the parameters of $v[f(wh, I), h, s]$ and $p(h, w, s)$. The intuition behind expression (3) is that the probability of a choice (h, w, s) can be expressed as the relative attractiveness—weighted by a measure of availability $p(h, w, s)$ —of jobs of type (h, w, s) . Given convenient parametric specifications of the functions v and p , the 78 parameters of the model can be estimated by maximizing the likelihood function formed on the basis of expression (3). The estimation is based on 1994 data collected by the 1995 Norwegian Survey of Level of Living, which includes detailed income data from tax reported records.⁶ We have restricted the ages of the individuals to between 20 and 62 in order to minimize the inclusion in the sample of individuals who, in principle, are eligible for retirement, because an analysis of retirement decisions is beyond the scope of this study. Moreover, the self-employed, as well as individuals receiving permanent disability benefits, are excluded from the sample. The sample contains 1,842 couples, 309 single females, and 312 single males. The estimates are reported in Aaberge and Colombino (2011).

When interpreting the random utility model illustrated above, it is important to stress that, in the model, household members choose from a set of jobs (characterized by h , w , and other characteristics s and j), not just from jobs that differ in hours of work h . Thus, the responses of households include many dimensions: hours, wage rates, and non-pecuniary job characteristics. Theoretical optimal taxation models typically consider effort as the agents' choice variable. Effort does not coincide with hours of work; it might include searching for jobs of better quality, putting more effort into each hour of work, or even configuring reported incomes in a more favourable way in view of taxation. A related concept—taxable income—has been used, among others, by Feldstein (1995) and Gruber and Saez (2002). The idea is that in evaluating the effects of changes in taxes, one should not just look at hours of work (and participation), because household responses include many other dimensions. At least some of these multidimensional responses are reflected in taxable income. However, the structural empirical models of labour supply used for tax-reform evaluations have traditionally

⁵For the derivation of the choice density (3), see Aaberge *et al.* (1999). Note that density (3) can be considered as a special case of the more general framework developed by Dagsgvik (1994). A more specialized type of continuous multinomial logit was introduced by Ben-Akiva and Watanatada (1981).

⁶At the time of performing the exercise presented in this paper, the 1994 sample was chosen because of the relatively stable macroeconomic conditions.

considered hours of work as the sole choice variable, implicitly equating hours of work and effort. An exception has been provided by Bourguignon and Spadaro (2005), who under rather special assumptions have been able to impute to each agent an effort value. Our model does not strictly equate effort to hours of work, because households—as a response to a change in the tax system—might choose a new job that differs from the previous job, with respect not only to hours of work but also to wage rate and other job characteristics. However, while we account for the disutility of hours of work and choice of sector, we only implicitly account (through the random utility component) for the fact that the other dimensions of effort might also bear a utility cost. Therefore, we cannot claim that our model is completely consistent with the “effort-taxable income” approach. We return to this limitation in Section V.

Behavioural Implications

In this section, we illustrate some of the behavioural implication of the estimates. First, we report the wage elasticities of labour supply because they are useful for understanding and interpreting the optimal taxation results that are presented in Section IV. Second, because the model is used for a rather ambitious operation (i.e., computing optimal tax-transfer rules), we illustrate the prediction performance of the model with an out-of-sample exercise.

The wage elasticities reported in Table 1 are computed by means of stochastic simulation. Note that the households face exogenous opportunity joint density functions of h , w , and s . Because many individuals in this labour supply model of discrete choice will not react to small exogenous changes, the elasticities in Table 1 have been computed as an average of the percentage changes in labour supply from a 10 per cent increase in the means of the wage densities. Given the simulated responses of each individual, we aggregate these to compute the aggregate elasticities. We find that the overall wage elasticity is equal to 0.12, which suggests rather low behavioural responses. However, by looking behind the overall elasticity, the picture changes substantially. The major features of the estimated labour supply elasticities can be summarized as follows: (a) labour supply for married women is far more elastic than for married men; (b) individuals belonging to low-income households are much more elastic than individuals belonging to high-income households. As demonstrated, for example, by Røed and Strøm (2002) and Meghir and Phillips (2008), these results are consistent with the findings in many recent studies. The profile of the wage elasticity across the income deciles is related to the hours worked. Households belonging to the higher-income brackets, on average, participate more and work longer hours; other things being equal, this tends

Table 1. *Labour-supply elasticities with respect to wage for single females, single males, married females, and married males by deciles of household disposable income (Norway, 1994)*

Family status	Type of elasticity	Income decile under the 1994 tax system	Female elasticities		Male elasticities	
			Own wage elasticities	Cross elasticities	Own wage elasticities	Cross elasticities
Single females and males	Elasticity of the probability of participation	I	0.59		0.00	
		II	0.45		0.00	
		III–VIII	0.06		0.06	
		IX	0.00		0.00	
		X	0.00		0.00	
		All	0.12		0.04	
	Elasticity of the conditional expectation of total supply of hours	I	–0.17		0.77	
		II	–0.04		0.00	
		III–VIII	–0.08		–0.08	
		IX	–0.07		0.00	
		X	0.00		0.00	
		All	–0.09		–0.02	
	Elasticity of the unconditional expectation of total supply of hours	I	0.42		0.77	
		II	0.42		0.00	
		III–VIII	–0.02		–0.02	
		IX	–0.07		0.00	
		X	0.00		0.00	
		All	0.02		0.02	
Married/cohabiting females and males	Elasticity of the probability of participation	I	1.03	–0.28	0.90	–0.23
		II	0.35	–0.14	0.79	0.00
		III–VIII	0.14	–0.23	0.13	–0.10
		IX	0.12	–0.12	0.06	–0.06
		X	0.07	0.00	0.06	–0.19
		All	0.21	–0.19	0.23	–0.11
	Elasticity of the conditional expectation of total supply of hours	I	1.51	–0.01	0.87	0.11
		II	0.62	–0.53	0.38	–0.08
		III–VIII	0.27	–0.24	0.18	–0.14
		IX	0.08	–0.22	0.02	–0.09
		X	0.19	–0.10	–0.02	–0.23
		All	0.31	–0.25	0.16	–0.13
	Elasticity of the unconditional expectation of total supply of hours	I	2.54	–0.29	1.77	–0.12
		II	0.97	–0.67	1.17	–0.08
		III–VIII	0.41	–0.47	0.31	–0.24
		IX	0.20	–0.34	0.08	–0.14
		X	0.26	–0.10	0.05	–0.42
		All	0.52	–0.42	0.39	–0.23

to lower the wage elasticity of labour supply. Table 1 also reveals that cross-elasticities are relevant in many cases, which supports the importance of modelling joint household decisions.

In principle, elasticities such as those illustrated above might be used to compute optimal tax-transfer rules, as in Saez (2001, 2002), Immervol *et al.* (2007), and Blundell *et al.* (2009), for example. As explained in Section I, we do not think that this procedure is totally satisfactory, because of the possible inconsistencies between the assumptions adopted by the theoretical optimal taxation model and the assumptions adopted in producing

Table 2. *Current tax rule in Norway, as of 1994, for singles without children and for couples without children and with two wage earners*

Gross earnings (NOK 1994)	Tax
(0–17,000)	0
(17,000–24,709)	$0.25x - 4,250$
(24,709–28,250)	$0.078x$
(28,250–140,500)	$0.302x - 6,328$
(140,500–208,000)	$0.358x - 14,196$
(208,000–234,500)	$0.453x - 33,956$
(234,500–)	$0.495x - 43,804$

Notes: x denotes annual earnings. Taxes include the part of social security contributions paid by the employee. 10,000 NOK $\simeq$ 1,250 euros.

the elasticities. For example, the empirical exercises, such as those mentioned above, typically adopt theoretical results that ignore cross-elasticities and income effects.⁷ Microeconomic models are based on assumptions that are much more flexible and general than those leading to the theoretical results. The approach we adopt in this paper exploits this greater flexibility, and guarantees consistency in the assumptions by obtaining the optimal tax-transfer rules computationally. That is, we iteratively run the microeconomic model of household behaviour until the social welfare function is maximized under the constraint that the total net tax revenue remains constant.

For the out-of-sample prediction exercise, we have used the model estimated on the 1994 sample and the 2001 data (exogenous variables) from the 2002 Norwegian Survey of Level of Living, in order to predict the choices made in 2001 under the new 2001 tax rules.⁸ Tables 2 and 3 describe some of the characteristics of the 1994 and 2001 tax regimes.

The basic features of the 1994 Norwegian tax system were determined by the major tax reform of 1992, which introduced a so-called dual income tax system characterized by a 28 per cent flat tax rate on capital income, in combination with progressive tax rates on labour income plus 7.8 per cent social insurance contribution. Further measures broadened the tax base of business income substantially, and removed various previous tax credits and deductions. In order to reduce incentives for taxpayers to classify labour income as capital income, the reform established rules for mandatory income splitting for dividing business income into capital and labour income,

⁷Aaberge and Colombino (2006, 2011) have reported on more detailed results for wage and income elasticities. Income effects are not negligible; their order of magnitude is close to the order of magnitude of wage elasticities and they vary substantially across gender, household type, location in the income distribution, and composition of income.

⁸Both 1994 and 2001 data are characterized by relatively stable macroeconomic conditions.

Table 3. *The 2001 tax function for singles without children and for couples without children and with two wage earners*

Gross earnings (NOK 2001)	Tax
[0–22,200)	0
[22,200–32,267)	$0.25x - 5,550$
[32,267–60,600)	$0.078x$
[60,600–144,545)	$0.358x - 16,968$
[144,545–183,182)	$0.296x - 8,064$
[183,182–289,000)	$0.358x - 19,348$
[289,000–793,200)	$0.493x - 58,363$
[793,200–)	$0.553x - 105,955$

Notes: x denotes annual earnings. Taxes include the part of social security contributions paid by the employee. 10,000 NOK $\simeq$ 1,250 euros.

Table 4. *Observed and predicted relative distributions of disposable income in 2001, as mean decile income in per cent of mean income*

Deciles	Couples		Single females		Single males	
	Observed	Simulated	Observed	Simulated	Observed	Simulated
1	50	49	45	47	41	42
2	68	64	56	61	54	55
3	77	74	68	71	65	67
4	83	83	79	79	76	76
5	89	90	90	88	87	86
6	95	98	101	98	97	97
7	102	107	111	108	107	108
8	111	117	123	121	119	121
9	125	131	139	138	137	141
10	199	187	189	188	218	207

with the resulting imputed wage income taxed according to a two-bracket progressive surtax. The associated top MTRs for wage earners and owners of small businesses were 49.5 and 52.4 per cent in 1994. Between 1992 and 2001, marginal rates, as well as the threshold of the highest bracket of the surtax, increased, resulting in the statutory tax rates for 2001 shown in Table 3. Disposable income is the variable used for comparing predicted outcomes to observed outcomes. The predictions are obtained individual by individual, evaluating the utility function—including the stochastic component drawn from the type III extreme value distribution—at each alternative, and identifying the selected alternative as the one with the highest utility level. The individual predictions are then aggregated into the 10 means of the 10 income deciles. Table 4 reports the results of an out-of-sample prediction exercise. The model is rather successful at reproducing the income distribution.

III. Design of Optimal Income Taxes

Framework of the Social Planner

The body of literature on optimal taxation relies on the maximization of social welfare functions, defined as summary measures of the distribution of individual utility levels, where utility levels are assumed to be interpersonally comparable. In this section, we explain the method adopted for allowing the aggregation of comparable individual welfare levels and the computational procedure used to solve the optimal taxation problem.

Individual Welfare Functions

Because the microeconomic labour supply model used in this study allows heterogeneous preferences for leisure and consumption and, moreover, because some individuals are singles and others live as couples, it makes no sense to treat the estimated utility functions as comparable individual welfare functions. Thus, it is necessary to introduce measures of individual welfare that permit interpersonal comparisons.⁹ Here, we follow an approach similar to the approaches advocated by Deaton and Muellbauer (1980), King (1983), and Hammond (1991). Specifically, we use a common utility function to evaluate the bundles chosen by the households according to their own preferences.¹⁰ The common utility function (or individual welfare function) is to be interpreted just as the input of a social welfare function. It is not used to simulate behaviour, only to evaluate—in a comparable way—the results of choices made according to the actual individual utility functions.

The individual welfare function V used by the social planner is specified as

$$\log V(y, h) = \gamma_2 \left(\frac{y^{\gamma_1} - 1}{\gamma_1} \right) + \gamma_4 \left(\frac{L^{\gamma_3} - 1}{\gamma_3} \right). \quad (4)$$

Here, L is leisure, defined as $L = 1 - (h/0.8736)$, and y is the individual's equivalent after tax income, defined by

$$y = \begin{cases} c = f(wh, I) & \text{for singles} \\ \frac{c}{\sqrt{2}} = \frac{1}{\sqrt{2}} f(w_F h_F, w_M h_M, I) & \text{for married/cohabiting individuals} \end{cases} \quad (5)$$

⁹See Boadway *et al.* (2002) and Fleurbaey and Maniquet (2006) for a discussion of interpersonal comparability of utility when preferences for leisure differ between individuals.

¹⁰An alternative approach has been recently proposed and illustrated by Decoster and Haan (2010).

Table 5. *Estimates of the parameters of the individual welfare function (Norway, 1994)*

Variable	Parameter	Estimate	Std dev.
<i>Income after tax (y)</i>	γ_1	-0.649	0.086
	γ_2	3.026	0.138
<i>Leisure (L)</i>	γ_3	-12.262	0.556
	γ_4	0.045	0.011

where F and M denote female and male respectively. The Box–Cox functional form of expression (4) is the same as that adopted for specifying the utility functions in the microeconomic model (Aaberge and Colombino, 2006, 2011). By dividing the couple income by $\sqrt{2}$, we transform the incomes of couples into comparable single individual incomes.¹¹ In order to estimate the parameters of the individual welfare function (4), we use expression (3) with the systematic part of the utility function (v) replaced by the individual welfare function (V) and conditional on the estimated opportunity densities p . The sample is the same as that used for estimating the model of Section II. Table 5 displays the parameter estimates of V .

A different way to circumvent the interpersonal comparability problem is to avoid interpersonal welfare level comparisons altogether, and to base the social evaluation exclusively on ordinal comparisons. We provide an example of this method in Table 11, where we present the number of “winners” under the optimal tax rules.¹²

Social Welfare Functions

When evaluating the distribution of the individual welfare effects of a tax system and/or a tax reform, it is necessary to summarize the gains and losses using a social welfare function. The simplest welfare function is the one that adds up the comparable welfare gains over individuals. The objection to the linear additive welfare function is that the individuals are given equal welfare weights, irrespective of whether they are poor or rich. However, concern for distributive justice requires that poor individuals

¹¹The square root scale is one of the equivalence scales commonly used in OECD publications. The number of household members, including children, is taken into account in the specification of the utility function, where it affects the marginal utilities of income and leisure (Aaberge and Colombino 2006, 2011).

¹²This is just an illustration; a proper application of the ordinal criterion would require a different definition of the optimal tax (e.g., the tax rule that maximizes the number of winners).

are assigned larger welfare weights than rich individuals. This structure is captured by the family of rank-dependent welfare functions,¹³

$$W = \int_0^1 q(t)F^{-1}(t)dt, \quad (6)$$

where F^{-1} is the left inverse of the cumulative distribution function of the individual welfare levels V with mean μ , and $q(t)$ is a positive weight-function defined on the unit interval. The social welfare functions (6) can be given a normative justification similar to the one underlying the expected utility social welfare functions introduced by Atkinson (1970). Given suitable continuity and dominance assumptions for the preference ordering $\succeq$ defined on the family of distributions F , Yaari (1987, 1988) has demonstrated that the following axiom characterizes the family of rank-dependent measures of social welfare functions (6), where $q(t)$ is a positive non-decreasing function of t .

Axiom (Dual Independence). *Let F_1 , F_2 , and F_3 be members of F and let $\alpha \in [0, 1]$. Then, $F_1 \succeq F_2$ implies $[\alpha F_1^{-1} + (1 - \alpha)F_3^{-1}]^{-1} \succeq [\alpha F_2^{-1} + (1 - \alpha)F_3^{-1}]^{-1}$.*

We refer to Yaari (1987, 1988) for a discussion of the difference between the dual independence axiom and the conventional independence axiom used to justify the expected utility social welfare functions. In this paper, we use the following specification of $q(t)$:

$$q_k(t) = \begin{cases} -\log t, & k = 1 \\ \frac{k}{k-1} (1 - t^{k-1}), & k = 2, 3, \dots \end{cases} \quad (7)$$

Note that the inequality aversion exhibited by the social welfare function W_k (associated with $q_k(t)$) decreases with increasing k . As $k \rightarrow \infty$, W_k approaches inequality neutrality and coincides with the linear additive welfare function defined by

$$W_\infty = \int_0^1 F^{-1}(t)dt = \mu. \quad (8)$$

It follows, by straightforward calculations, that $W_k \leq \mu$ for all k and that W_k is equal to the mean μ for finite k if and only if F is the egalitarian distribution. Thus, W_k can be interpreted as the equally distributed individual welfare level. As recognized by Yaari (1988), this property suggests

¹³Several other authors have discussed the rationale for rank-dependent measures of inequality and social welfare (e.g., Sen, 1974; Hey and Lambert, 1980; Donaldson and Weymark, 1980, 1983; Weymark, 1981; Ben Porath and Gilboa, 1994; Aaberge, 2001).

Table 6. *Distributional weight profiles of four different social welfare functions*

	W_1 (Bonferroni)	W_2 (Gini)	W_3	W_∞ (Utilitarian)
$q(0.01)/q(0.5)$	6.64	1.98	1.33	1
$q(0.05)/q(0.5)$	4.32	1.90	1.33	1
$q(0.30)/q(0.5)$	1.74	1.40	1.21	1
$q(0.95)/q(0.5)$	0.07	0.10	0.13	1

that C_k , defined by

$$C_k = 1 - \frac{W_k}{\mu}, \quad k = 1, 2, \dots, \quad (9)$$

can be used as a summary measure of inequality.¹⁴ As noted by Aaberge (2000, 2007), C_1 is actually equivalent to the measure of inequality that was proposed by Bonferroni (1930), whilst C_2 is the Gini coefficient. Aaberge (2000, 2007) has demonstrated that C_1 exhibits a strong downside inequality aversion, and is particularly sensitive to changes that concern the poor part of the population, whilst C_2 normally pays more attention to changes that take place in the middle part of the income distribution. The C_3 coefficient exhibits upside inequality aversion, and is thus particularly sensitive to changes that occur in the upper part of the income distribution. Because of the close relationship between C_1 , C_2 , and C_3 Aaberge (2007) has proposed treating them as a group, and calling them the Gini nuclear family of inequality measures. In order to ease the interpretation of the inequality aversion profiles exhibited by W_1 , W_2 , W_3 , and W_∞ , Table 6 provides the ratios of the corresponding weights—defined by equation (7)—of the median individual and the 1 per cent poorest, the 5 per cent poorest, the 30 per cent poorest, and the 5 per cent richest individuals for different social welfare criteria. As can be observed from the weight profiles provided by Table 6, W_1 will be particularly sensitive to changes in policies that affect the welfare of the poor, whereas the inequality aversion profile of W_3 is rather moderate, and W_∞ exhibits neutrality with respect to inequality.

Optimal Taxation Problem

We strictly consider only personal income taxation. Following the tradition of the optimal income tax literature, all the other dimensions of the wider

¹⁴Note that Aaberge (2001) has provided an axiomatic justification for using the C_k measures as the criteria for ranking Lorenz curves. Thus, the justification of the social welfare function $W_k = \mu(1 - C_k)$, defined by equation (6) (and equation (9)), can also be made in terms of a value judgement of the trade-off between the mean and (in)equality in the distribution of welfare.

tax system (VAT, consumption taxes, payroll taxes, social assistance, etc.) have been kept constant, as of 1994 in Norway. The optimal taxation problem considered in this exercise can be formulated as

$$\begin{aligned}
 & \max_{\vartheta} W[V(y_{1F}, h_{1F}), V(y_{2F}, h_{2F}), \dots, V(y_{nF}, h_{nF}), \\
 & \quad V(y_{1M}, h_{1M}), V(y_{2M}, h_{2M}), \dots, V(y_{nM}, h_{nM})] \\
 & \text{s.t.} \\
 & y_{iF} = y_{iM} = c_i / \sqrt{2}, \quad i = 1, 2, \dots, n \\
 & \sum_{i=1}^n [w_{iF}h_{iF} + w_{iM}h_{iM} + I_i - f(w_{iF}h_{iF}, w_{iM}h_{iM}, I_i; \vartheta)] \geq G \\
 & (c_i, h_{iF}, h_{iM}, s_{iF}, s_{iM}, j_i) = \arg \max_{(w, h, s, j) \in B_i} U_i(c, h_F, h_M, s_F, s_M, j) \text{ s.t. } c_i \\
 & \quad = f(w_{iF}h_{iF}, w_{iM}h_{iM}, I_i; \vartheta), \quad i = 1, 2, \dots, n.
 \end{aligned} \tag{10}$$

Here, n is the number of households, and G denotes the total net tax revenue required (set equal to the current one in our exercise). For simplicity of exposition, expression (10) assumes that n households are couples, whereas, in fact, we consider both couples and singles. In expression (10), each couple contributes to the social welfare functions, with two terms corresponding to the individual welfare functions of the two partners. For singles, we have just one term $V(y_i, h_i)$ and $y_i = c_i$ (according to expression (5)). All the variables are the same as those that appear in expression (1). The function $c_i = f(w_{iF}h_{iF}, w_{iM}h_{iM}, I_i; \vartheta)$, which transforms gross incomes $(w_{iF}h_{iF}, w_{iM}h_{iM}, I_i)$ into net available income c_i , denotes a class of tax rules defined up to a vector of parameters ϑ . We consider a class of piecewise-linear tax rules with a (positive or negative) lump-sum transfer and five income brackets. Therefore, the parameters are the amount of the lump-sum transfer, the lower and upper limits of the income brackets, and the MTR applied to the income brackets. Household i maximizes its own utility given the tax rule $f(w_{iF}h_{iF}, w_{iM}h_{iM}, I_i; \vartheta)$ by choosing the job $(c_i, h_{iF}, h_{iM}, s_{iF}, s_{iM}, j_i)$. Taking the individual utility-maximizing choices into account as a constraint (i.e., the incentive-compatibility constraint), the social planner searches for the tax rule (i.e., the parameter vector ϑ) that maximizes the social welfare function W , subject to the constraint that the total net tax revenue must at least be as large as G . The social welfare function W takes as arguments the evaluations of the chosen jobs, according to the individual welfare function V . Given the very flexible and general specifications adopted for the random utility functions and of the opportunity sets (Aaberge and Colombino, 2011), problem (10) cannot be solved analytically. The maximization of W is performed by a global maximization procedure that efficiently scans the parameter space. At each run of

the iterative procedure, the maximization of the individual utility function is simulated by the microeconomic model described in Section II. The search for the optimal tax rule is limited to the class of piecewise-linear rules, with five brackets:

$$c = \begin{cases} z + d - \tau_1 z & \text{if } 0 < z \leq z_1; \\ z + d - \sum_{i=1}^{l-1} \tau_i (z_i - z_{i-1}) - \tau_l (z - z_{l-1}) & \\ \text{if } z_{l-1} < z \leq z_l, \quad l = 2, 3, 4; \\ z + d - \sum_{i=1}^4 \tau_i (z_i - z_{i-1}) - \tau_5 (z - z_4) & \text{if } z > z_4. \end{cases} \quad (11)$$

Here, c is the net available income, z is the sum of gross market income (earnings plus capital income) and taxable public transfers, $(\tau_1, \tau_2, \dots, \tau_5)$ are the MTRs applied to the five income brackets, z_i is the upper limit of the i th bracket ($i = 1, 2, 3$, and 4), $z_0 = 0$ and d is a lump sum that can be positive (i.e., a lump-sum transfer) or negative (i.e., a lump-sum tax). Thus, each particular tax rule is characterized by 10 parameters $(\tau_1, \tau_2, \tau_3, \tau_4, \tau_5, z_1, z_2, z_3, z_4, d)$. The tax rule is quite flexible because the MTRs are allowed to be positive or negative (less than 1 in absolute value), and the bracket limits are allowed to take any positive value only subject to the constraints $z_i \geq z_{i-1}$. The tax rule specified by expression (11) replaces the current rule, as of 1994, whose main characteristics are illustrated by the examples of Table 2, and also belongs to the class of piecewise linear tax rules.¹⁵ The dataset is the same as the one used for the estimation of the model (Section II). The identification of the optimal tax rules consists of the following four steps.

- (a) For each household, we simulate the opportunity set, which contains the observed job plus 199 market and non-market alternatives drawn from the estimated p densities defined in Section II. For each household and each alternative in the opportunity set, we then draw a value ε from the type III extreme value distribution. Next, the new tax rule is applied to the gross incomes of the individual earners in order to obtain disposable incomes (income after tax) corresponding to each alternative in the choice set. For each household, a new choice $(c, h_F, h_M, s_F, s_M, j)$ for couples or (c, h, s, j) for singles—is given by the alternative that

¹⁵Taxes include the part of social security contributions paid by the employee.

Table 7. *Optimal tax rules according to alternative social welfare criteria*

	Social welfare function			
	W_1 (Bonferroni)	W_2 (Gini)	W_3	W_∞ (Utilitarian)
τ_1	-0.30	-0.80	-0.70	-0.80
τ_2	0.06	0.20	0.22	0.24
τ_3	0.29	0.26	0.26	0.29
τ_4	0.39	0.38	0.37	0.33
τ_5	1.00	1.00	1.00	1.00
d	-13,600	-7,500	-5,200	-5,800
z_1	10,000	10,000	10,000	10,000
z_2	120,000	130,000	140,000	230,000
z_3	220,000	230,000	240,000	290,000
z_4	730,000	720,000	720,000	790,000

Notes: d , z_1 , z_2 , z_3 , and z_4 are in 1994 NOK (10,000 NOK $\simeq$ 1,250 euros).

- maximizes the household-specific utility functions defined by assumption (2).¹⁶
- (b) To each decision maker (wife, husband, or single), an equivalent income y is imputed according to expression (5). The purpose of this procedure is to convert the distribution of incomes (c) across heterogeneous families into a distribution of (equivalent) incomes (y) across adult individuals.
- (c) As a result of the previous steps, now for each individual we have a simulated pair (y , h). Then, we compute the individual welfare levels by applying to the chosen (y , h) the individual welfare function (4).
- (d) Then, we compute the social welfare function W_k for $k = 1, 2, 3, \infty$.

The optimization is performed by iterating steps (a)–(d) in order to find the tax rule in the class (11) that produces the highest value of W_k for each value of k , under the constraint of constant total net tax revenue.¹⁷

IV. Optimal Tax-Transfer Schedules

The results of our exercise are reported in Tables 7–11. Table 7 displays the optimal tax rules. In order to make it easier to compare the behavioural responses to the 1994 tax system and to the various optimal tax systems, we

¹⁶Colombino (1998), Colombino *et al.* (2010), and Blundell and Shephard (2012) have used a different method, where the maximum utility is not found by simulation but is measured instead by the expected maximum utility (McFadden, 1978). The two methods are asymptotically equivalent, but the method adopted in this paper turns out to be more flexible and robust for producing disaggregated results.

¹⁷The optimal tax-transfer parameters are determined by an iterative grid-search procedure developed by Tom Wennemo at the Research Department of Statistics Norway. Each optimization requires the evaluation of approximately 200,000 tax-transfer rules.

Table 8. *Percentage of individuals by income intervals under different tax systems*

Income intervals	Proportions located in various gross income segments			
	Couples (males)	Couples (females)	Single males	Single females
1994 tax system				
0–30,000	5	16	0	0
30,000–130,000	11	33	26	24
130,000–230,000	31	35	41	51
230,000–730,000	52	16	33	24
730,000–>	2	0	0	0
W_1-optimal tax system				
0–30 000	2	10	0	0
30,000–130,000	9	32	22	22
130,000–230,000	29	41	42	51
230,000–730,000	58	17	35	27
730,000–>	2	0	0	0
W_2-optimal tax system				
0–30,000	3	12	0	0
30,000–130,000	9	32	23	22
130,000–230,000	28	39	41	50
230,000–730,000	59	17	36	28
730,000–>	1	0	0	0
W_3-optimal tax system				
0–30,000	3	12	0	0
30,000–130,000	8	31	23	21
130,000–230,000	27	39	41	50
230,000–730,000	60	17	36	28
730,000–>	2	0	0	0
W_∞-optimal tax system				
0–30,000	3	13	0	0
30,000–130,000	8	31	22	20
130,000–230,000	25	37	40	50
230,000–730,000	62	18	38	30
730,000–>	2	0	0	0

report the proportions of individuals by family status in specific tax income brackets in Table 8. Tables 9 and 10 provide additional information about the behavioural implications of the optimal tax rules. Table 11 displays the percentages of winners under the optimal rule by income deciles of the 1994 income distribution.

Under any social welfare function, the MTRs are continuously increasing for all levels of income. Clearly, the pattern of elasticities—sharply decreasing with respect to income—illustrated in Table 1 contributes to the profile of the optimal MTR. The most striking results are represented by the negative MTR on the first bracket and by the 100 per cent MTR on the last bracket. These results are obviously driven by the pattern of the elasticities displayed in Table 1. From each of the panels of Table 1, we

Table 9. *Percentage changes in participation rates, annual hours of work, and disposable income under the optimal tax rules*

		Social welfare function			
		W_1 (Bonferroni)	W_2 (Gini)	W_3	W_∞ (Utilitarian)
Single males	Participation rates	2.3	2.3	2.3	2.3
	Annual hours	4.9	5.0	5.1	6.0
	Disposable income	10.0	9.7	10.0	11.9
Single females	Participation rates	4.0	4.4	4.4	4.8
	Annual hours	6.0	6.6	6.6	9.0
	Disposable income	4.7	4.6	4.5	6.6
Couples	Participation rates, M	2.6	2.3	2.3	2.7
	Participation rates, F	5.8	4.3	3.9	3.3
	Annual hours, M	5.9	6.2	6.6	9.4
	Annual hours, F	10.6	8.1	7.0	6.3
	Disposable income	9.2	9.4	9.9	13.3

Table 10. *Percentage changes in labour supply (total hours) by household income decile under the optimal tax rules*

		Social welfare function							
		W_1 (Bonferroni)		W_2 (Gini)		W_3		W_∞ (Utilitarian)	
Income decile under the 1994 system		Male	Female	Male	Female	Male	Female	Male	Female
Singles	I	60.5	64.7	57.3	54.4	62.8	54.4	62.8	60.3
	II	18.6	17.9	18.6	21.3	18.6	21.3	20.3	29.3
	III–VIII	0.9	3.0	1.2	4.5	1.1	4.5	1.7	7.2
	IX	0.0	0.0	0.0	0.0	0.0	0.0	2.6	−0.4
	X	1.3	0.0	1.3	0.0	1.3	0.0	1.3	0.0
	All	4.9	6.0	5.0	6.6	5.1	6.6	6.0	9.0
Couples	I	50.4	74.4	43.8	60.7	42.8	56.0	49.5	60.8
	II	22.2	22.9	23.3	19.5	22.9	20.5	32.7	18.6
	III–VIII	2.6	7.7	3.5	5.1	4.2	3.6	6.7	2.7
	IX	0.7	0.5	0.7	1.0	0.7	0.5	1.2	−0.3
	X	−3.3	0.0	−2.9	0.2	−2.9	0.9	−1.4	0.1
	All	5.9	10.6	6.2	8.1	6.6	7.0	9.4	6.3

observe that the labour supply of the 10 per cent poorest is very responsive to changes in economic incentives, whereas the 10 per cent richest are inelastic. Moreover, by comparing the fourth and fifth panels of Table 1, for married/cohabitating females we see that hours supplied (given participation), in particular for those belonging to the poorest couples, are far more responsive than participation.¹⁸ A negative MTR on low incomes – infact,

¹⁸Except when income effects are assumed to be zero, the relationship between the optimal MTR and the wage elasticities is complicated. Other things being equal, a large (small)

Table 11. *Percentage of winners under optimal tax rules*

Income decile under the 1994 system		Social welfare function							
		W_1 (Bonferroni)		W_2 (Gini)		W_3		W_∞ (Utilitarian)	
		Male	Female	Male	Female	Male	Female	Male	Female
Singles	I	79	76	83	72	83	72	79	69
	II	66	62	66	55	62	55	55	55
	III–VIII	86	68	85	68	81	68	77	66
	IX	79	45	83	45	83	45	83	48
	X	76	34	79	38	79	38	86	41
	All	82	63	82	62	79	62	76	61
Couples	I	62	64	64	67	62	65	61	64
	II	70	72	72	73	73	73	70	73
	III–VIII	84	85	84	86	84	87	83	87
	IX	85	87	86	88	88	90	88	91
	X	71	69	72	70	74	72	79	78
	All	79	80	70	81	80	82	79	83

a subsidy or a tax credit on the wage rate—is close to policies actually implemented, such as the Working Families Tax Credit in the UK, the Earned Income Tax Credit in the US, and the In-Work Tax Credit in Sweden. The 100 per cent MTR on the last bracket—despite the fact that it could hardly be realistically implemented—does make sense within the limits of our model. As shown in Table 1, people in the richest decile exhibit, on average, a wage elasticity of labour supply equal, or very close, to zero (with the exception of married women). Some individuals from this segment of the population (1.5 per cent) are willing to work the same amount of hours, despite a reduction of the net wage rate, and the non-pecuniary characteristics of the job (captured by the utility random component) might induce them to choose jobs where the marginal net wage is zero (i.e., their earnings become a rent that, as such, is captured by the optimal tax rule).¹⁹ This argument carries over *a fortiori* to unearned income (completely inelastic in our model), which for this segment of the population might represent a very significant part of total income. The overall picture emerging from our exercise is in sharp contrast with most of the results

compensated wage elasticity contributes to a low (high) optimal MTR. Table 1 reports uncompensated wage elasticities. However, even when accounting for the incomes elasticities (reported by Aaberge and Colombino, 2011), the resulting compensated wage elasticities are inversely related to the income levels.

¹⁹An anonymous referee correctly argued that this same segment of the population might be willing to accept a MTR above 100 per cent (i.e., a negative marginal net wage rate). However, we decided to constrain the optimal MTRs to be less than or equal to 100 per cent, because official rates above 100 per cent would not be realistically considered.

obtained by the numerical exercises based on the optimal tax formulae of Mirrlees. The typical outcome of these exercises envisages a positive lump-sum transfer, which is progressively taxed away by very high MTRs on lower incomes (i.e., a negative income tax mechanism); after the income level where the transfer is exhausted, the tax rule is close to proportional. However, Tuomala (2010) has suggested that these results are essentially forced by the restrictive assumption typically made upon preferences, elasticities, and distribution of productivities (or wage rates).

Table 7 shows that the more egalitarian the social welfare criterion, the more progressive the optimal tax rule. For example, the optimal rule according to Bonferroni is more progressive than the optimal rule according to Gini, which, in turn, is more progressive than the optimal utilitarian rule.

The lump sum d turns out to be a tax. This result can be explained by the fact that households with low or average/incomes are particularly sensitive to changes in marginal taxes (see Table 1). Thus, the MTRs on low and average incomes are kept low, both for minimizing distortions and for fulfilling distributive goals. However, because the total net tax revenue must be kept unchanged, the optimal tax rule envisages a universal lump-sum tax. A possible practical implementation close to a lump-sum tax might be represented by a tax on wealth or on property (e.g., on owner-occupied houses). According to this interpretation, the optimal tax rules would imply—with respect to the 1994 rule—a lower taxation on earnings complemented by a property tax.

All the optimal rules imply a higher income after tax for most levels of gross income (Table 9). In other words, the optimal rules are able to extract the same total tax revenue from a larger total gross income (i.e., under a lower average tax rate). Together with the optimal MTRs of Table 7, this result provides a controversial perspective in view of the tax reforms implemented in many developed countries during the last decades. In most cases, these reforms have embodied the idea of improving efficiency and labour-supply incentives through a lower average tax rate and lower MTRs on the highest incomes.²⁰ Our results give clear support to the first part (i.e., lowering the average tax rate). As for the second part, the picture is less clear-cut. Our results suggest that a lower average tax rate should be mainly obtained by lowering the marginal and average tax rates, particularly on low and average incomes (and also on a substantial part of high incomes), and by sharply increasing them on very high income levels.²¹

²⁰For example, Blundell (1996) has reported that, during the 1980s and early 1990s, in some countries, the top MTRs were cut from 70–80 per cent down to about 40–50 per cent. Concerning these issues, the discussion by Røed and Strøm (2001) is especially relevant.

²¹A second important difference between our exercise and the implemented reforms referred to in the main text is that these reforms typically envisaged a reduction of the total tax

Table 10 shows that the strongest labour-supply response comes from households in the lower income deciles (i.e., those who show a more elastic labour supply). While females in couples receive a stronger incentive to work under the Bonferroni regime than under the Utilitarian regime, the opposite is the case for the males. This is because the wife faces, on average, lower wages than the husband, and the lower tax brackets are more relevant for her (i.e., those where the Bonferroni regime imposes much lower MTRs than the Utilitarian – and the current – regime). However, the Utilitarian regime is especially favourable (also compared to the current regime) for those who decide to locate themselves in high tax brackets, where husbands are more likely to be found. The implication is that a more egalitarian criterion also involves stronger work incentives for married women (and especially those in the lower income deciles), and therefore also a more egalitarian intergender distribution of income.

Table 11 shows the percentage of winners under the optimal rules, by marital status, gender, and household income decile under the current 1994 rule. Individuals are defined as winners if their welfare is higher under the new tax rule than under the current 1994 rule. All the optimal rules would largely “win the referendum” against the current rule, because they all imply a strong majority of winners. However, the percentage of winners varies substantially across the different subgroups, and especially across income deciles. Single women in the IX and X income deciles are the only ones who would “vote against” all the optimal tax rules. The current (1994) tax system provides important deductions that favour, in particular, the group of relatively well-off single women with children. The deductions are removed in the class of tax-transfer rule we optimize upon. As a consequence, a majority of these women turn out to be losers under the optimal rules.

V. Conclusions

We have performed an exercise in designing optimal income taxes. Unlike what is typically done in the literature, in this exercise we do not rely on *a priori* theoretical optimal taxation results, but instead we employ a microeconomic model of labour supply in order to maximize a social welfare function with respect to a parametrically defined income tax rule. Modern microeconomic models of labour supply are based on very general and flexible assumptions. They can accommodate many realistic features, such as general structures of heterogeneous preferences, simultaneous decisions of household members, complicated (non-convex, non-continuous,

revenue, together with a reduction in the average tax rate, while in our simulations we keep the total tax revenue unchanged.

non-differentiable, etc.) constraints and opportunity sets, multidimensional heterogeneity of both households and jobs, quantitative constraints, etc. It is simply not feasible (at least, so far) to obtain analytical solutions for the optimal income taxation problem in such environments. Yet, these features are very relevant and important, especially in view of evaluating or designing reforms. Analytical solutions remain indispensable for understanding the grammar of the problem, and for suggesting promising classes of tax-transfer systems that can then be more deeply investigated with the microeconomic model.²² The microeconomic model adopted in this paper, and fully described by Aaberge and Colombino (2006, 2011), is designed to allow for a detailed description of complex choice sets and budget constraints. The model is used to identify, by simulation, the tax rule that maximizes a social welfare function. We have kept fixed the current (1994) system of transfers, income support, and social assistance policies, but we allow for a lump sum that can be positive (i.e., a transfer) or negative (i.e., a tax). We have explored a variety of different social welfare criteria. The MTRs always turn out to be monotonically increasing with income. More egalitarian social welfare functions tend to imply more progressive tax rules. For all the social welfare functions used, the optimal bottom MTR is negative and the optimal top MRT always turns out to be 100 per cent for sufficiently high gross income levels (depending on the social welfare function, approximately above 720,000–790,000 NOK 1994), which concerns not more than 2 per cent of the tax payers. The negative MTR on the lowest income bracket suggests a mechanism close to policies such as the Working Families Tax Credit in the UK or the Earned Income Tax Credit in the USA. The 100 per cent top MTR can be mainly explained by the inelastic labour supply at the top of the income distribution (Table 1) and by non-pecuniary characteristics that might make a job attractive, even though it carries a 100 per cent MTR. All the optimal tax rules imply an average tax rate lower than the current 1994 one, and imply—with respect to the current rule—lower marginal rates on low and/or average income levels, and a higher marginal rate on very high income levels. The pattern of wage elasticities of labour supply illustrated in Section II helps to explain the profile of the optimal tax rules. Our results are partially at odds with the tax reforms that have taken place in many countries during the last decades. While these reforms have embodied the idea of lowering average tax rates, the way in which they have been implemented has typically

²²The philosophy inspiring this approach is similar to the one adopted long ago in engineering and also, recently and successfully, in other applications of mechanism design (auctions, negotiation procedures, matching markets, etc.) where analytical solutions are complemented by computational simulations or experiments that account for a host of realistic features, which cannot be included in the theoretical model. Roth (2002) has provided a very inspired survey of this approach.

consisted of reducing the top marginal rates. In contrast, our results suggest lowering average tax rates by reducing marginal rates, except for those of very high income levels.

Even though we think that the approach illustrated here can usefully complement theoretical work and analytical solutions, and actually improve upon them concerning the representation of preferences, constraints, and policies, clearly there are many dimensions of the tax-transfer rules that are relevant for their evaluation (e.g., implementation and administrative costs) but are beyond the purpose of our exercise. Moreover, some of the results illustrated in Section IV might change with the inclusion in the behavioural model of features that are currently not taken into account. A candidate for further refinements is the modelling of the choice by households at the top of the income distribution. For example, the optimal top MTR might turn out to be lower than 100 per cent if we were able to fully account for other dimensions of households' responses, such as intercountry mobility and taxable income response (see Feldstein, 1995; Gruber and Saez, 2002).²³

References

- Aaberge, R. (2000), Characterizations of Lorenz Curves and Income Distributions, *Social Choice and Welfare* 17, 639–653.
- Aaberge, R. (2001), Axiomatic Characterization of the Gini Coefficient and Lorenz Curve Orderings, *Journal of Economic Theory* 101, 115–132.
- Aaberge, R. (2007), Gini's Nuclear Family, *Journal of Economic Inequality* 5, 305–322.
- Aaberge, R. and Colombino, U. (2006), Designing Optimal Taxes with a Microeconomic Model of Household Labour Supply, IZA Discussion Paper No. 2468.
- Aaberge, R. and Colombino, U. (2011), Empirical Optimal Income Taxation: A Microeconomic Application to Norway, CHILD Working Paper No. 16/2011.
- Aaberge, R., Colombino, U., and Strøm, S. (1999), Labour Supply in Italy: An Empirical Analysis of Joint Household Decisions, with Taxes and Quantity Constraints, *Journal of Applied Econometrics* 14, 403–422.
- Aaberge, R., Colombino, U., and Strøm, S. (2000), Labour Supply Responses and Welfare Effects from Replacing Current Tax Rules by a Flat Tax: Empirical Evidence from Italy, Norway and Sweden, *Journal of Population Economics* 13, 595–621.
- Aaberge, R., Colombino, U., and Strøm, S. (2004), Do More Equal Slices Shrink the Cake? An Empirical Investigation of Tax-Transfer Reform Proposals in Italy, *Journal of Population Economics* 17, 767–785.
- Aaberge, R., Dagsvik, J. K., and Strøm, S. (1995), Labour Supply Responses and Welfare Effects of Tax Reforms, *Scandinavian Journal of Economics* 97, 635–659.
- Atkinson, A. B. (1970), On the Measurement of Inequality, *Journal of Economic Theory* 2, 244–263.

²³ However, based on previous exercises where we constrained the top MTR to be lower than 100 per cent, we expect the overall qualitative features of the optimal tax rule to remain unaffected.

- Ben-Akiva, M. and Watanatada, T. (1981), Application of a Continuous Spatial Choice Logit Model, in C. F. Manski and D. McFadden (eds.), *Structural Analysis of Discrete Data with Econometric Applications*, MIT Press, Cambridge, MA.
- Ben Porath, E. and Gilboa, I. (1994), Linear Measures, the Gini Index, and the Income-Equality Trade-Off, *Journal of Economic Theory* 64, 443–467.
- Blundell, R. (1996), Labour Supply and Taxation, in M. Devereux (ed.), *The Economics of Tax Policy*, Oxford University Press, Oxford.
- Blundell, R., Brewer, M., Haan, P., and Shephard, A. (2009), Optimal Income Taxation of Lone Mothers: An Empirical Comparison of the UK and Germany, *Economic Journal* 119, F101–F121.
- Blundell, R. and Shephard, A. (2012), Employment, Hours of Work and the Optimal Taxation of Low Income Families, *Review of Economic Studies* 79, 481–510.
- Boadway, R., Marchand, M., Pestieau, P., and Racionero, M. (2002), Optimal Redistribution with Heterogeneous Preferences for Leisure, *Journal of Public Economic Theory* 4, 475–498.
- Bonferroni, C. (1930), *Elementi di Statistica Generale*, Seeber, Firenze.
- Bourguignon F. and Spadaro, A. (2005), Tax-Benefit Revealed Social Preferences, PSE Working Paper 2005–22.
- Colombino, U. (1998), Evaluating the Effects of New Telephone Tariffs on Residential Users' Demand and Welfare. A Model for Italy, *Information Economics and Policy* 10, 283–303.
- Colombino, U., Locatelli, M., Narazani, E., and O'Donoghue, C. (2010), Alternative Basic Income Mechanisms: An Evaluation Exercise with a Microeconomic Model, *Basic Income Studies* 5 (1).
- Dagsvik, J. K. (1994), Discrete and Continuous Choice, Max-Stable Processes and Independence from Irrelevant Attributes, *Econometrica* 62, 1179–1205.
- Deaton, A. and Muellbauer, J. (1980), *Economics and Consumer Behavior*, Cambridge University Press, Cambridge.
- Decoster, A. and Haan, P. (2010), Empirical Welfare Analysis in Random Utility Models of Labour Supply, IZA Discussion Paper 5301.
- Diamond, P. (1980), Income Taxation with Fixed Hours of Work, *Journal of Public Economics* 12, 101–110.
- Diamond, P. (1998), Optimal Income Taxation: An Example with a U-Shaped pattern of Optimal Marginal Tax Rates, *American Economic Review* 88, 83–95.
- Donaldson, D. and Weymark, J. A. (1980), A Single Parameter Generalization of the Gini Indices of Inequality, *Journal of Economic Theory* 22, 67–86.
- Donaldson, D. and Weymark, J. A. (1983), Ethically Flexible Indices for Income Distributions in the Continuum, *Journal of Economic Theory* 29, 353–358.
- Feldstein, M. (1995), The Effect of Marginal Tax Rates on Taxable Income: A Panel Study of the 1986 Tax Reform Act, *Journal of Political Economy* 103, 551–572.
- Fleurbaey, M. and Maniquet, F. (2006), Fair Income Tax, *Review of Economic Studies* 73, 55–84.
- Fortin, B., Truchon, M., and Beauséjour, L. (1993), On Reforming the Welfare System: Workfare Meets the Negative Income Tax, *Journal of Public Economics* 31, 119–151.
- Gruber, J. and Saez, E. (2002), The Elasticity of Taxable Income: Evidence and Implications, *Journal of Public Economics* 84, 1–32.
- Hammond, P. J. (1991), Interpersonal Comparisons of Utility: Why and How They Are and Should Be Made?, in J. Elster and J. E. Roemer (eds.), *Interpersonal Comparisons of Well-Being. Studies in Rationality and Social Change*, Cambridge University Press, in collaboration with Maison des Sciences de l'Homme, Cambridge, 200–254.
- Haan, P. and Wrohlich, K. (2010), Optimal Taxation: The Design of Child-Related Cash and In-Kind Benefits, *German Economic Review* 11, 278–301.

- Hey, J. D. and Lambert, P. J. (1980), Relative Deprivation and the Gini Coefficient: Comment, *Quarterly Journal of Economics* 94, 567–573.
- King, M. (1983), Welfare Analysis of Tax Reforms Using Household Data, *Journal of Public Economics* 21, 183–214.
- Kleven, H. J., Kreine, C. T., and Saez, E. (2009), The Optimal Income Taxation of Couples, *Econometrica* 77, 537–560.
- Immervoll, H., Kleven, H. J., Kreiner, C. T., and Saez, E. (2007), Welfare Reforms in European Countries: A Microsimulation Analysis, *Economic Journal* 117, 1–44.
- McFadden, D. (1978), Modeling the Choice of Residential Location, in A. Karlqvist, L. Lundqvist, F. Snickars, and J. Weibull (eds.), *Spatial Interaction Theory and Planning Models*, North-Holland, Amsterdam, 75–96.
- Meghir, C. and Phillips, D. (2008), Labour Supply and Taxes, IZA Discussion Paper no. 3405.
- Mirrlees, J. A. (1971), An Exploration in the Theory of Optimal Income Taxation, *Review of Economic Studies* 38, 175–208.
- Revesz, J. T. (1989), The Optimal Taxation of Labour Income, *Public Finance* 44, 453–475.
- Roth, A. E. (2002), The Economist as Engineer: Game Theory, Experimental Economics and Computation as Tools of Design Economics, *Econometrica* 70, 1341–1378.
- Røed, K. and Strøm, S. (2002), Progressive Taxes and the Labour Market—Is the Trade-Off between Equality and Efficiency Inevitable?, *Journal of Economic Surveys* 16, 77–100.
- Saez, E. (2001), Using Elasticities to Derive Optimal Income Tax Rates, *Review of Economic Studies* 68, 205–229.
- Saez, E. (2002), Optimal Income Transfer Programs: Intensive versus Extensive Labour Supply Responses, *Quarterly Journal of Economics* 117, 1039–1073.
- Sen, A. (1974), Informational Bases of Alternative Welfare Approaches, *Journal of Public Economics* 3, 387–403.
- Tuomala, M. (1990), *Optimal Income Tax and Redistribution*, Clarendon Press, Oxford.
- Tuomala, M. (2010), On Optimal Non-Linear Income Taxation: Numerical Results Revisited, *International Tax and Public Finance* 17, 259–270.
- Weymark, J. (1981), Generalized Gini Inequality Indices, *Mathematical Social Sciences* 1, 409–430.
- Yaari, M. E. (1987), The Dual Theory of Choice under Risk, *Econometrica* 55, 95–115.
- Yaari, M. E. (1988), A Controversial Proposal Concerning Inequality Measurement, *Journal of Economic Theory* 44, 381–397.

First version submitted April 2010;
final version received November 2011.